



MFML: THE ULTIMATE EQUATION & INTUITION GUIDE

"Don't just memorize symbols. See the geometry in the matrices, the physics in the gradients, and the boundaries in the constraints."

1. 🏠 MATRICES, SPACES & SVD

Vector Spaces & Independence

- **Linear Indep:** $\sum c_i v_i = 0 \implies c_i = 0 \forall i$.
- **Basis:** A set of linearly independent vectors that span the space. **Dimension** = # of basis vectors.
- **Rank(A):** Max # of independent columns/rows. If an $n \times n$ matrix has Rank $< n$, it is singular ($\det = 0$).

Eigenvectors (v) & Eigenvalues (λ)

Vectors that only scale (don't rotate) under transform A:

$$Av = \lambda v \implies \det(A - \lambda I) = 0$$

Singular Value Decomposition (SVD)

Factorizes any $m \times n$ matrix A into rotations and scales:

$$A = U \Sigma V^T$$

- **U** ($m \times m$): Orthogonal left singular vectors ($U^T U = I$). Columns are eigenvectors of AA^T .
- **V** ($n \times n$): Orthogonal right singular vectors ($V^T V = I$). Columns are eigenvectors of $A^T A$.
- **Σ** ($m \times n$): Diagonal matrix of singular values $\sigma_i = \sqrt{\lambda_i(A^T A)}$.

2. 🏔️ GRADIENTS, JACOBIAN & HESSIAN

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (scalar) and $g: \mathbb{R}^n \rightarrow \mathbb{R}^m$ (vector).

- **Gradient (∇f):** Vector of 1st partials (Steepest Ascent).

$$\nabla f(x) = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]^T \in \mathbb{R}^{n \times 1}$$

- **Jacobian (J):** Matrix of 1st partials for vector outputs.

$$J_{ij} = \frac{\partial g_i}{\partial x_j} \implies J = \begin{bmatrix} \nabla g_1(x)^T \\ \vdots \\ \nabla g_m(x)^T \end{bmatrix} \in \mathbb{R}^{m \times n}$$

- **Hessian (H or $\nabla^2 f$):** Matrix of 2nd partials (Curvature).

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j} \in \mathbb{R}^{n \times n} \quad (H \text{ is symmetric!})$$

Maxima, Minima & Saddle Points

- Find critical points where $\nabla f(x) = 0$.
- Check Eigenvalues (λ_i) of Hessian **H** at that point:
 - All $\lambda_i > 0 \implies H$ is **Positive Definite** ($H \succ 0$) $\rightarrow$ **Local Min** $\cup$
 - All $\lambda_i < 0 \implies H$ is **Negative Definite** ($H \prec 0$) $\rightarrow$ **Local Max** $\cap$
 - Mixed $+/- \implies H$ is **Indefinite** $\rightarrow$ **Saddle Point**

3. ⚙️ GRADIENT DESCENT & VARIANTS

Let $g_t = \nabla f(x_t)$ be gradient at step t . η = Learning Rate.

Standard GD: Walk strictly downhill:

$$x_{t+1} = x_t - \eta g_t$$

Momentum: Adds past velocity to power through saddles:

$$v_t = \gamma v_{t-1} + \eta g_t \implies x_{t+1} = x_t - v_t$$

AdaGrad: Adapts η per parameter (ϵ prevents div by 0):

$$G_t = G_{t-1} + g_t \odot g_t \implies x_{t+1} = x_t - \frac{\eta}{\sqrt{G_t + \epsilon}} \odot g_t$$

RMSProp: Fixes AdaGrad's "dying η " with Exp. Moving Average:

$$E_t = \rho E_{t-1} + (1-\rho)(g_t \odot g_t) \implies x_{t+1} = x_t - \frac{\eta}{\sqrt{E_t + \epsilon}} \odot g_t$$

Adam (Momentum + RMSProp):

Maintains EMA of gradients (m_t) and squared gradients (v_t):

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t \quad (1st \text{ Moment})$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (g_t \odot g_t) \quad (2nd \text{ Moment})$$

Bias Correction: $\hat{m}_t = m_t / (1 - \beta_1^t)$ and $\hat{v}_t = v_t / (1 - \beta_2^t)$

$$\text{Update: } x_{t+1} = x_t - \frac{\eta}{\sqrt{\hat{v}_t + \epsilon}} \odot \hat{m}_t$$

4. 📦 CONSTRAINED OPTIMIZATION & DUALITY

Goal: $\min f(x)$ subject to $g_i(x) \leq 0$ and $h_j(x) = 0$.

The Lagrangian ($\mathcal{L}$): Penalizes constraint violations.

$$\mathcal{L}(x, \alpha, \beta) = f(x) + \sum_i \alpha_i g_i(x) + \sum_j \beta_j h_j(x)$$

Min-Max Equality & Duality

- **Primal (p^*):** $p^* = \min_x \max_{\alpha \geq 0, \beta} \mathcal{L}(x, \alpha, \beta)$
- **Dual (d^*):** $d^* = \max_{\alpha \geq 0, \beta} \min_x \mathcal{L}(x, \alpha, \beta)$
- **Weak Duality:** $d^* \leq p^*$. **Strong Duality:** $d^* = p^*$
(Holds if convex & Slater's condition is met. Gap = 0!).

Karush-Kuhn-Tucker (KKT) Conditions

If Strong Duality holds, the optimal points MUST satisfy these 4:

1. **Stationarity:** $\nabla_x \mathcal{L}(x^*, \alpha^*, \beta^*) = 0$
2. **Primal Feasibility:** $g_i(x^*) \leq 0, h_j(x^*) = 0$
3. **Dual Feasibility:** $\alpha_i^* \geq 0$
4. **Comp. Slackness:** $\alpha_i^* g_i(x^*) = 0$ (Either constraint is tight $g = 0$, or multiplier $\alpha = 0$)

5. ⚙️ PCA & SVM FORMULATIONS

Principal Component Analysis (PCA)

Objective: Find unit vector w maximizing variance of projected data.

$$\max_w w^T \Sigma w \quad \text{subject to} \quad w^T w = 1$$

- $\Sigma = \frac{1}{n} X^T X$ is the covariance matrix of *mean-centered* data.
- **Solution:** By Lagrangian, $\Sigma w = \lambda w$. Optimal w is the Eigenvector with the largest Eigenvalue λ .

Support Vector Machine (SVM)

Goal: Maximize Margin ($2/\|w\|$) $\equiv$ Minimize $\frac{1}{2}\|w\|^2$.

1. Primal Formulation (Soft Margin):

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad \text{s.t.} \quad y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0$$

Unconstrained Hinge Loss Form:

$$\min_{w, b} \sum_{i=1}^n \max(0, 1 - y_i(w^T x_i + b)) + \frac{\lambda}{2} \|w\|^2$$

2. Dual Formulation & Kernel Trick:

Derived via KKT. Depends ONLY on dot products $x_i^T x_j$!

$$\max_{\alpha} \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j (x_i^T x_j)$$

subject to: $\sum \alpha_i y_i = 0$ and $0 \leq \alpha_i \leq C$

- **Kernel Trick:** Replace $x_i^T x_j$ with $K(x_i, x_j)$ to map to infinite dimensions. (e.g., $K(x, z) = \exp(-\gamma \|x - z\|^2)$).
- **Support Vectors:** Any x_i where $\alpha_i > 0$. Only these define the boundary!

6. ☠️ FATAL EXAM TRAPS

- **Matrix Shapes:** Always write dimensions under equations! $A_{m \times n} B_{n \times p} = C_{m \times p}$. If inner dimensions don't match, STOP.
- **Gradient Ascent vs Descent:** If asked to MAXIMIZE, you must ADD the gradient ($+\eta \nabla f$). GD subtracts ($-\eta \nabla f$).
- **PCA Centering:** If you do not subtract the mean ($X - \mu$) before computing $X^T X$, your components are completely wrong.
- **Definiteness:** A matrix with all positive numbers $\neq$ Positive Definite. You MUST check if the *Eigenvalues* > 0 (or use Sylvester's Criterion).